

LYON-BRON AP, France

WMO: 074800

Lat: 45.7261N

Lon: 4.9378E

Elev: 201

StdP: 98.93

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-3.6	-9.7	1.7	-2.2	-7.2	2.1	-1.7	13.4	7.9	11.9	8.2	2.4	0	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.8	20.2	31.8	19.8	30.0	19.4	21.6	30.4	20.8	29.1	20.1	28.1	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	24.2	18.0	13.3	23.8	17.3	12.7	23.3	63.4	30.3	60.8	29.3	58.4	28.2	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.5	8.3	-7.6	36.3	2.5	2.2	-9.4	37.9	-10.8	39.2	-12.2	40.4	-14.0	42.0		
			-8.1	23.0	2.5	1.0	-9.9	23.6	-11.3	24.2	-12.7	24.8	-14.5	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	3.9	5.0	8.9	12.4	16.2	20.5	22.6	21.8	18.1	13.8	8.1	4.8
	DBStd	7.55	3.97	3.99	3.60	3.40	3.57	3.83	3.41	3.44	3.28	3.81	3.95	4.09
	HDD10.0	674	192	142	64	14	2	0	0	0	0	11	82	168
	HDD18.3	2323	447	372	293	178	85	20	5	6	43	146	307	421
	CDD10.0	1792	3	3	29	87	194	316	390	367	243	129	25	6
	CDD18.3	400	0	0	0	2	19	86	137	115	36	5	0	0
	CDH23.3	3814	0	0	0	27	185	829	1353	1110	291	19	0	0
	CDH26.7	1420	0	0	0	2	35	303	550	461	70	0	0	0
Wind	WSAvg	3.2	3.1	3.3	3.5	3.5	3.3	3.4	3.2	2.9	2.9	3.0	3.1	3.0
Precipitation	PrecAvg	852	54	44	48	73	80	69	68	65	92	104	101	54
	PrecMax	1125	135	108	144	193	178	163	152	156	254	263	281	158
	PrecMin	656	16	6	6	10	17	26	21	26	22	18	32	10
	PrecStd	127	28	25	31	43	37	32	33	31	59	50	58	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	16.8	21.4	26.3	29.8	35.1	36.2	37.3	31.0	25.4	19.5	16.2
		MCWB	11.2	10.1	12.5	15.7	18.5	21.1	20.4	20.2	18.9	17.1	14.0	11.9
	2%	DB	13.1	14.6	19.1	23.5	27.4	32.1	33.8	33.8	28.9	23.5	17.6	14.2
		MCWB	9.7	9.5	11.3	14.3	17.1	20.1	20.3	20.1	18.7	16.6	13.3	10.7
	5%	DB	11.6	12.8	17.2	21.4	25.4	30.1	31.7	31.5	26.8	21.7	16.0	12.8
		MCWB	8.9	8.5	10.7	13.3	16.5	19.4	20.0	19.6	17.8	16.0	12.4	9.9
	10%	DB	10.0	11.2	15.4	19.4	23.4	28.1	29.9	29.1	24.8	20.0	14.3	11.0
		MCWB	7.9	7.8	10.0	12.2	15.5	18.7	19.4	19.2	17.2	15.3	11.3	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.9	13.5	16.2	19.7	22.8	22.5	22.1	20.9	18.4	14.7	12.2
		MCDB	14.6	15.1	19.1	24.4	27.4	30.9	31.9	31.5	27.7	23.1	18.2	15.0
	2%	WB	10.1	9.9	12.2	14.9	18.2	21.4	21.5	21.2	19.5	17.4	13.6	11.1
		MCDB	12.5	13.7	17.1	22.3	25.6	29.5	30.8	29.8	26.0	21.9	16.8	13.9
	5%	WB	9.1	9.0	11.4	13.8	17.2	20.2	20.8	20.5	18.7	16.5	12.6	10.0
		MCDB	11.3	12.3	16.1	20.3	23.7	28.1	29.4	28.8	24.9	20.6	15.6	12.4
	10%	WB	7.9	8.0	10.5	12.7	16.1	19.3	20.0	19.7	17.9	15.6	11.6	8.8
		MCDB	9.9	11.0	14.8	18.3	21.7	26.7	28.1	27.4	23.5	19.3	14.1	10.7
Mean Daily Temperature Range	5% DB	MDBR	5.9	7.5	9.4	10.0	10.2	10.9	10.9	11.3	10.5	8.6	6.7	5.9
		MCDBR	8.1	10.8	12.8	13.8	13.6	13.8	13.9	15.2	13.9	10.8	9.2	8.1
		MCWBR	5.5	6.6	6.6	6.5	5.7	5.2	4.7	5.1	5.5	5.3	5.5	5.4
	5% WB	MCDBR	7.6	9.8	11.2	12.8	12.1	12.8	12.5	13.2	11.5	9.6	8.3	7.6
		MCWBR	5.3	6.3	6.1	6.4	5.5	5.2	4.7	5.0	4.9	5.1	5.4	5.4
Clear-Sky Solar Irradiance	taub	0.311	0.332	0.362	0.398	0.403	0.405	0.402	0.396	0.378	0.364	0.331	0.309	
	taud	2.474	2.411	2.346	2.250	2.284	2.315	2.321	2.348	2.379	2.419	2.468	2.479	
	Ebn at Noon	797	845	864	859	865	863	860	852	836	794	762	758	
	Edh at Noon	68	87	107	129	129	125	123	115	103	86	69	62	
All-Sky Solar Radiation	RadAvg	1.25	2.10	3.40	4.52	5.28	6.29	6.16	5.32	4.06	2.54	1.41	1.05	
	RadStd	0.15	0.30	0.42	0.58	0.50	0.50	0.52	0.40	0.32	0.25	0.16	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page